

I/O Architectures

input/output as a subsystem of components that moves coded data between external devices and a host system (CPU and main memory)

- ▶ Blocks of main memory that are devoted to I/O functions
- ▶ Buses that provide the means of moving data into and out of the system
- ▶ Control modules in the host and in peripheral devices
- ▶ Interfaces to external components such as keyboards and disks
- ▶ Cabling or communications links between the host system and its peripherals

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Printer

- The manner in which control its I/O.
 - Programmed I/O
 - Interrupt-driven I/O
 - Memory-mapped I/O
 - Direct
 - Channel-attached I/O

- Channel-attached i/o

Programmed I/O (1/2)

- ▶ Polled I/O (or port I/O)
- ▶ The CPU continually monitors (polls) a control register associated with each I/O port.
- ▶ When a byte arrives in the port, a bit in the control register is also set. The CPU eventually polls the port and notices that the "data ready" control bit is set
- ▶ The CPU resets the control bit, retrieves the byte, and processes it according to instructions programmed for that particular port
- ▶ When the processing is complete, the CPU resumes polling the control registers as before.

- Otal para dispositivos muy lentos (Teclados)
- Cnótico con dispositivos muy rápidos (Discos)

Programmed I/O (2/2)

- ▶ We have programmatic control over the behavior of each device
- ▶ The CPU is in a continual “busy wait” loop until it starts servicing an I/O request
- ▶ Another problem is in deciding how frequently to poll; some devices might need to be polled more frequently than others

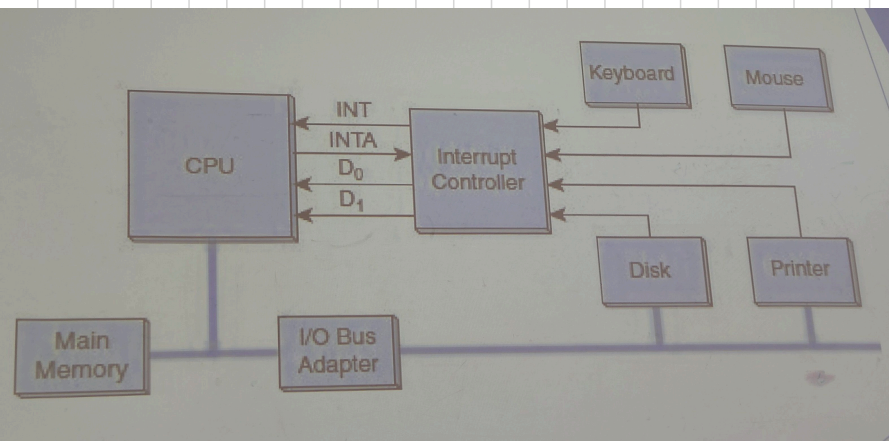
- una pregunta y esperu...

Interrupted-driven I/O

- Dice cuando acaba, avisa
- Opuesto al programado

con una interrupción

- Oposto al programando



Memory-mapped I/O

- Comparte misma dirección del espacio de memoria
- Cada dispositivo I/O tiene su propio bloque de memoria
- Transferencia de datos de I/O a dispositivo I/O requiere mover bytes de la dirección de memoria al otro dispositivo.
- Podemos usar el mismo set de instrucciones para mover datos de la I/O y memoria, simplificando el diseño.

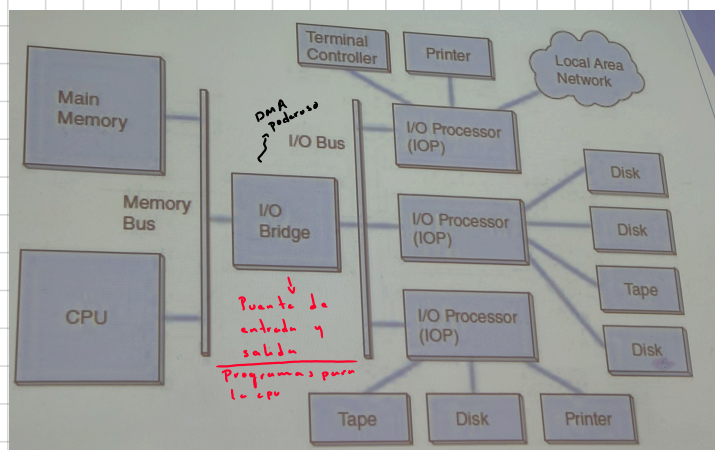
Program I/O $\rightarrow$ **Muy lento**
 Interrupted I/O $\rightarrow$ **Más rápido**
 Memory I/O $\rightarrow$ **Impresoras y así.**

Direct memory access (DMA)

- La CPU se desentanda del I/O.
- La CPU le pasa al controlador de DMA la ubicación de los bytes.

Channel attached - I/O:

- La mayoría de sistemas lo usan.
- Tradicionalmente usado en mainframe computers.



- Mayor escalamiento: Si queremos más entradas-salidas los añadimos.
- No es tan frecuente.

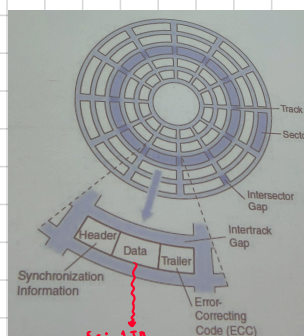
Storage technologies

- Magnetic Disk
- RAID
- Optical Disk
- Magnetic tape

Magnetic Disk Technology

- Disk drives are called random (sometimes direct) access devices
- A circular surface (called platter) was coated on both sides with a magnetic oxide
- Each unit of storage on a disk, the sector, has a unique address that can be accessed independently of the sectors around it.
- Sectors are divisions of concentric circles called tracks.
- On most systems, every track contains exactly the same number of sectors.
- Each sector contains the same number of bytes.
- The data is written more "densely" at the center of the disk than at the outer edge.

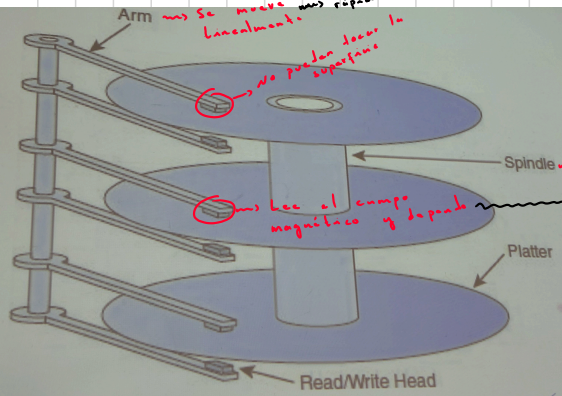
The platter



- El cuanto se puede almacenar en un disco depende del track más pequeño.
- Si el track es pequeño más espacio para almacenar.

750 GB - Los otros 40 GB es para Headers y trailer

Discos están al vacío !!



- Gira $\rightarrow$ cambia la velocidad del giro $\rightarrow$ más datos al espacio para esto!
- Para escribir: De electromagnetismo a electricidad
- Para leer: De electricidad a electromagnetismo

El 50, segundos:

Terminology

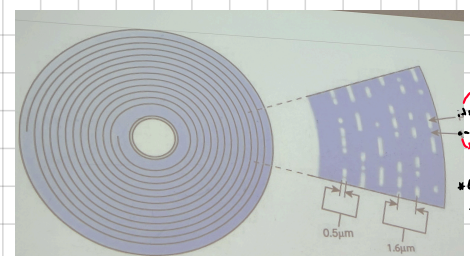
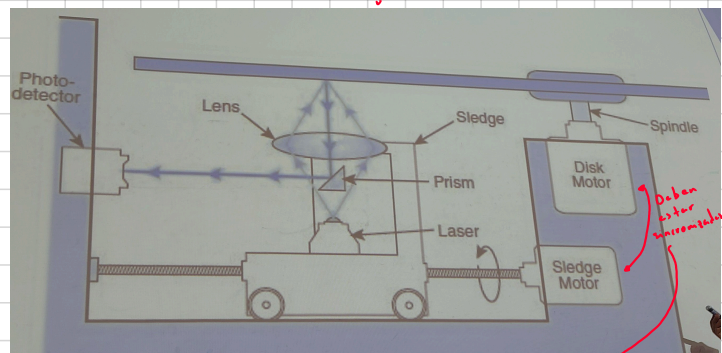
- Seek time is the time it takes for a disk arm to position itself over the required track.
- Rotational delay is the time it takes for the required sector to position itself under a read/write head. *(Cuanto tiempo hace a volar) una promesa de*
- Access time is the sum of the rotational delay and seek time. *recorrer mucho track*
- Transfer time is the access time plus the time it takes to actually read the data from the disk. *Promesa de cuanto me demora en llegar a un sector en específico*
- Transfer time varies depending on how much data is read.
- Latency is a measure of the amount of time it takes for the desired sector to move beneath the read/write head after the disk arm has positioned itself over the desired track *cuanto me demora de verdad en el transfer de datos*
- Latency is a direct function of rotational speed

Optical Disks

- CD-ROM (compact disc-read only memory), which can hold more than 0.5GB of data
- CD-ROMs are a read-only medium, making them ideal for software and data distribution
- CD-R (CD-recordable) *no se puede*
- CD-RW (CD-rewritable)
- WORM (write once read many) disks are optical storage devices often used for long-term data archiving and high-volume data output.
- For long-term archival storage of data, some computer systems send output directly to optical storage rather than paper or microfiche. This is called computer output laser disc (COLD).
- Robotic storage libraries called optical jukeboxes provide direct access to myriad optical disks.
- Jukeboxes can store dozens to hundreds of disks, for total capacities of 50GB to 100GB and beyond

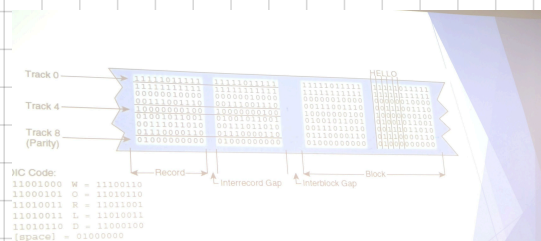
CD-ROM

- CD-ROMs are polycarbonate (plastic) disks 120mm (4.8 inches) in diameter to which a reflective aluminum film is applied.
- The aluminum film is sealed with a protective acrylic coating to prevent abrasion and corrosion.
- The aluminum layer reflects light that emits from a green laser diode situated beneath the disk.
- The reflected light passes through a prism, which diverts the light into a photodetector.
- The photodetector converts pulses of light into electrical signals, which it sends to decoder electronics in the drive



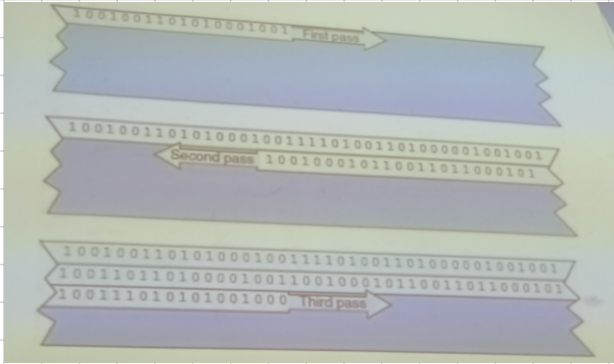
Magnetic Tape

- A cellulose-acetate film one-half inch wide (1.25cm) was coated on one side with a magnetic oxide.
- Twelve hundred feet of this material were wound onto a reel, which then could be hand-threaded on a tape drive
- Nine Track Format: Data was written across the tape one byte at a time, creating one track for each bit.
- An additional track was added for parity, making the tape nine tracks wide
- Serpentine: Instead of the bytes being perpendicular to the edges of the tape, as in the nine-track format, they are written "lengthwise," with each byte aligning in parallel with the edge of the tape.
- A stream of data is written along the length of the tape until the end is reached; then the tape reverses and the next track is written beneath the first one



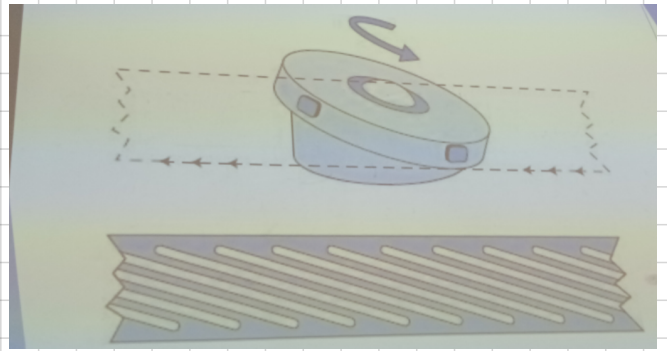
*Problemas de espacio !!

Se crea la serpiente



Pasamos a DAT

Serpentina



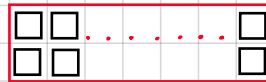
LTO: Linear Tape Open:

- Estándar
- Usa compresión, hasta 2:1 (2,8 TB a 4 TB)

RAID:

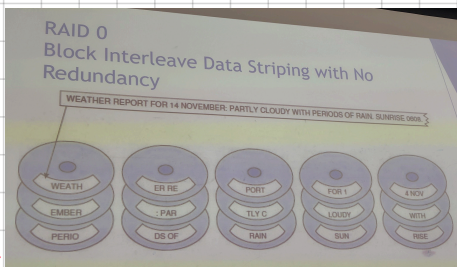
- Single large expensive disks (SLEDs)
- Redundant Arrays of "Inexpensive" Disks *→ Baratos y, cuando a falta*
- Redundant Arrays of independent Disks
 - Computer sees a single disk
- Levels:

RAID + Hot Swap Disk *→ Se puede sacar el disco en uso.*

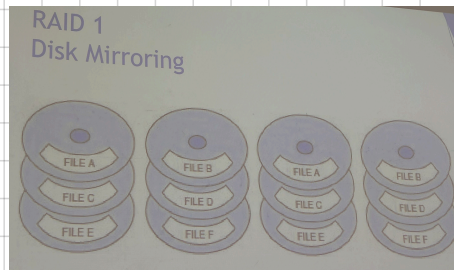


→ Se pone y se quita corriendo, evita cortes

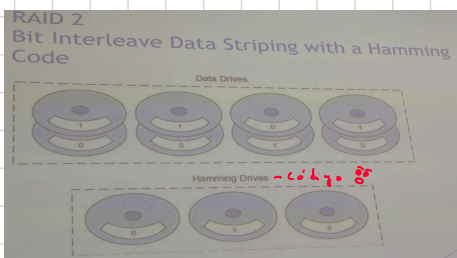
RAID 0 - RAID 6 *→ Hay de RAID 0-2-6*, RAID 1+0 *→ Combinación, es el "8"*



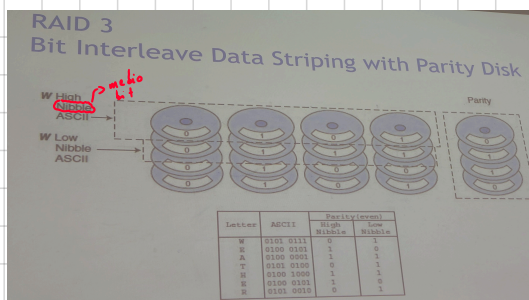
- Va escribiendo pedacito a pedacito.
- Es bueno si se quiere aprovechar el disco.
- Malos: Si daña un disco, vamos *→ Si es lento, es recuperable, si son bits parciales*
- Si se pierden datos



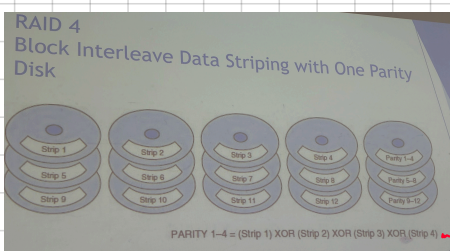
- Se escribe en disco y en la paragona
- Reconstruye la copia
- Datos siempre disponible
- Sincronización lenta
- Disco muy bajito de que pierda datos.
- Problema: Costoso y se necesita el doble.



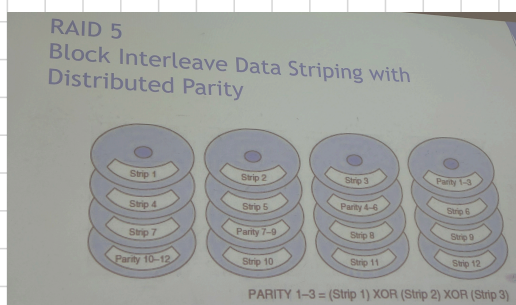
- Parámetros
- El código de Hamming está distribuido por todos los bytes.
- Casi imposible perder datos.
- DEMASIADO lento *→ Sería el ideal, pero muy complejo.*



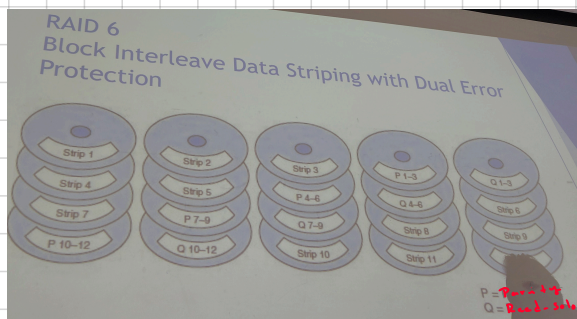
- Se pierde un 50%
- Muy problemático en cuanto la reconstrucción de bits y bytes
- No fue implementado



- Verifican que los datos están bien
- RAID 0 + paridad
- 20% disco se va en paridad *→ No es un 20%*



- Se sabe que hay errores, no se puede corregir.
- Va siguiendo con la paridad *→ Se guardan para recuperar.*



y se como 50% disco

* RAID 1 to 5:

- Cada dato tiene su copia
- No asegura discos
- Dependiendo de la tier, lo guarda donde quieras,
- Se pierde menos información